

Computational Supplement: 1 + 3 Covariant CMB Anisotropies

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v1.0.0 — First public analytical reproducibility release
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1 Scientific scope

This supplement records the analytical companion to the associated paper. The bundle is an independent Python reconstruction, from first 1 + 3 covariant principles, of the almost-Friedmann–Lemaître results of Gebbie, Dunsby & Ellis, *Annals of Physics* **282**, 321–394 (2000), doi:10.1006/aphy.2000.6034. It is archived at doi:10.25375/uct.34069509, active on publication of v1.0.0.

It is written bottom-up: not a port of, and not a wrapper on, an existing Boltzmann code, and it inherits no other code’s conventions. It reproduces; it does not argue. No new effect is claimed.

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2 Theory-to-computation map

Table 1 links each published equation the numerics implement to its provenance and to the .py file and object implementing it, together with the test that checks it. Table 2 gives the notation and the names the quantities carry in the code.

Both are *generated* from config/implementation-register.toml by scripts/make_tables.py, and tests/test_register.py resolves every module, object and test named in them. They cannot drift from the code: a rename breaks the build rather than quietly falsifying a table printed here.

The harmonic normalisation is $Q_{A_\ell} = (-k_{\text{phys}})^{-\ell} D_{(A_\ell)} Q$. Both candidate conventions are run end to end; only this one reproduces the real, opposite-sign form of the external hierarchies.¹

Nomenclature is kept strictly separate: τ_ℓ are **mode** coefficients and τ_{A_ℓ} are **multipole** coefficients. *Annals II* notes that the distinction is not made in Bardeen-variable treatments. Every released figure plots a multipole mean-square, which holds for general geometries, rather than a mode mean-square, which holds only for almost-Robertson–Walker ones.

¹*Annals I* prints $(-1)^\ell$ at the corresponding step where its own definition gives i^ℓ . The covariant multipole τ_{A_ℓ} is invariant under the choice, so the misprint never produced a wrong result; the mode coefficients are not invariant, and the two statements are distinct. Both are asserted as tests. A misprint, recorded in provenance/ANNALS-II-ANTECEDENT-AND-CORRECTIONS-v1.md.

Table 1: Equations implemented in the numerics, with their provenance and the code that implements them. Equation numbers are those of the *published* papers. Rows with no code are not yet implemented.

Quantity	Source	Equation	File	Object	Checked by
$\beta_\ell = O^{A_\ell} O_{A_\ell} = (\ell!)^2 2^\ell / (2\ell)!$	Annals I	(24)	functions/harmonics/weights.py	beta	test_beta_module_matches_symbolic_exactly
$\beta_\ell = \frac{\ell}{2\ell-1} \beta_{\ell-1}, \beta_{\ell+1} = \frac{\ell+1}{2\ell+1} \beta_\ell$	Annals I	(25)	functions/harmonics/weights.py	beta	test_beta_recurrence_descending_general_ell
$\Delta_\ell = 4\pi\beta_\ell/(2\ell+1)$	Annals I	(119)	functions/harmonics/weights.py	delta_over_pi	—
α_ℓ , defined by $\beta_\ell = \alpha_\ell^{-1}(2\ell+1)$	Annals II	App. F, p. 380	functions/harmonics/weights.py	alpha	test_f4_is_direct_substitution_into_f3
free-streaming weight $w_\ell = (\ell+1)^2/[(2\ell+3)(2\ell+1)]$	Annals II	(F.1), p. 379	functions/harmonics/weights.py	free_streaming_weight	test_f1_weight_is_the_mode_recursion_weight_shifted
covariant recursion $-\dot{\tau}_\ell = (k_{\text{com}}/a)[w_\ell \tau_{\ell+1} - \tau_{\ell-1}]$	Annals II	(F.1), p. 379	functions/harmonics/free_streaming.py	free_stream	test_the_three_normalisations_are_one_solution
β -normalised recursion, conformal time	Annals II	(F.3), p. 380	functions/harmonics/free_streaming.py	free_stream	test_the_three_normalisations_are_one_solution
α -normalised recursion	Annals II	(F.4), p. 380	functions/harmonics/free_streaming.py	free_stream	test_f4_variable_is_exactly_the_spherical_bessel_function
free-streaming solution $\alpha_\ell^{-1} \tau_\ell = j_\ell(k_{\text{com}} \Delta \eta)$	Annals II	App. F, p. 380	functions/harmonics/free_streaming.py	analytic_projection	test_f4_variable_is_exactly_the_spherical_bessel_function
$\dot{\Theta}_\ell = k[\frac{\ell}{2\ell-1} \Theta_{\ell-1} - \frac{\ell+1}{2\ell+3} \Theta_{\ell+1}]$	Hu & Sugiyama	(6), p. 2601	functions/harmonics/external.py	WEIGHTS	test_external_hierarchies_match_the_bundle_at_machine_precision
$\dot{F}_{\nu\ell} = \frac{k}{2\ell+1} [F_{\nu(\ell-1)} - (\ell+1)F_{\nu(\ell+1)}]$	Ma & Bertschinger	(49), (50)	functions/harmonics/external.py	WEIGHTS	test_external_hierarchies_match_the_bundle_at_machine_precision
$\dot{\Delta}_{T\ell} = \frac{k}{2\ell+1} [\ell \Delta_{T(\ell-1)} - (\ell+1) \Delta_{T(\ell+1)}]$	Seljak & Zaldarriaga	(3d)	functions/harmonics/external.py	WEIGHTS	test_external_hierarchies_match_the_bundle_at_machine_precision
$\int_0^\infty z^{-m} j_\ell^2(z) dz$, corrected to carry $[(m/2)!]^2$	Annals II	(199), p. 372	scripts/derive_ell_range.py	bessel_integral_corrected	test_annals_ii_199_needs_the_square_in_its_denominator

Quantity	Source	Equation	File	Object	Checked by
$\ell_A = \pi \Delta \eta_*/r_s$, bounding criterion 5's range	Annals II	§8.3.2, p. 372	<code>scripts/derive_ell_range.py</code>	<code>acoustic_scale</code>	—
S_P , S_{DISW} , S_{ISW} — the temperature source split	Annals II	(177), (178), (179), p. 366	—	—	—
integral solution, $\beta_\ell \tau_\ell(\eta_0)/(2\ell + 1)$ projected through j_ℓ	Annals II	(176), p. 365	—	—	—
truncation condition: four consecutive vanishing harmonics force $\sigma_{ab} = 0$	Annals II	App. E, (E.1), p. 378	—	—	—

Table 2: Notation, and the names the quantities carry in the code. The mode/multipole distinction is the one that matters most: *Annals II* notes that it is not made in Bardeen-variable treatments.

Symbol	In code	Meaning	Note
$\tau(x, e)$	—	temperature anisotropy as a function of position and direction	—
τ_{A_ℓ}	—	multipole coefficients — the covariant objects	valid for general geometries; what every released figure plots
τ_ℓ	tau, values	mode coefficients	almost-Robertson–Walker only. Conflating these two is how a sign is lost
O^{A_ℓ}	—	$e^{(A_\ell)}$, the PSTF basis on the sphere	Annals I (15)
Q_{A_ℓ}	—	harmonic basis tensor, $(-k_{\text{phys}})^{-\ell} D_{(A_\ell)} Q$	the adopted normalisation; see criterion 3
β_ℓ	beta	$O^{A_\ell} O_{A_\ell}$	Hu & Sugiyama’s normalisation
α_ℓ	alpha	$(2\ell + 1)/\beta_\ell$	Ma & Bertschinger’s and Seljak & Zaldarriaga’s normalisation
k_{com}	k_com	comoving wavenumber	never a bare k in this codebase
k_{phys}	k_phys	physical wavenumber, k_{com}/a	Appendix F’s k is comoving; the conventions sheet’s is physical
η	eta	conformal time, $dt = a d\eta$	—
$\Delta\eta_*$	d_eta	$\eta_0 - \eta_*$	—
j_ℓ	spherical_jn	spherical Bessel function	the free-streaming solution
u^a	—	the threading 4-velocity	carried generic; frames are specialisations made at the end
σ_{ab}	—	shear	σ is the shear and nothing else
A_a	—	acceleration, $\dot{u}_a$	—
Θ_ℓ	—	Hu & Sugiyama’s multipole variable	β -normalised

3 The acceptance spine

Appendix F of the antecedent, pp. 379–380, Eqs. (F.1)–(F.4), links the covariant mode recursion to external treatments written in real coefficients. Reproducing that chain tests the harmonic normalisation without introducing a representation, and it is where this bundle’s independence lies.²

Three external treatments are used — Hu & Sugiyama Eq. (6), Ma & Bertschinger Eqs. (49) and (50), and Seljak & Zaldarriaga Eq. (3d) — each transcribed from its own paper. Hu & Sugiyama write the recursion in the β normalisation and the other two in the α normalisation, which is why Appendix F prints both (F.3) and (F.4). A basis phase leaking into the couplings would break at least one of the two matches.

Wilson is not reproduced: a coefficient transcribed through Appendix F would be the same source as Appendix F rather than an independent fourth, the paper is pre-arXiv, and Appendix F names Wilson & Silk while keying Wilson.

4 Frames and specialisation

The solution is carried with a **generic** u^a . The explicit frames appear as choices made at the end, not assumed at the start: the Newtonian frame $\tilde{\sigma}_{ab} = 0$, which is the specialisation *Annals II* itself makes at

²The comparison applies the $(2\ell + 1)^{-1}$ division that the text preceding (F.4) specifies and the display omits; (F.4) is otherwise correct as printed. A misprint, recorded.

p. 365 in order to remove the shear terms, and the energy frame $q_a = 0$. Carrying the generic threading and specialising late is the difference the 1 + 3 covariant approach buys; a gauge-fixed treatment fixes the frame before the hierarchy is solved and cannot exhibit it.

This is also what fixes the status of the next release. Because v1.0.0 is the general case, **v1.5.0’s restricted case is a recovery and a check** — it must return v1.0.0’s numbers, and that requirement is the regression test rather than a new result.

Two consequences are visible in the figures. The decomposition of the temperature source into primary, Doppler and integrated terms (§7.1.1, Eqs. (177)–(179)) is frame-dependent while the total is not, so it is drawn in two frames. And footnote 29 on p. 365 asserts that the Newtonian choice also removes the high- ℓ truncation difficulty of Appendix E; that is a checkable claim, and the convergence figure checks it in both frames.

5 Comparison and its range

The C_ℓ comparison is Sachs–Wolfe only: (196) descends from (181), in which the acoustic modulation $\cos(kr_s)$ is dropped by taking $r_s^* \rightarrow 0$, and §8.3.2 sets the transfer function to unity. Its accuracy is therefore bounded by the term that was dropped. The range is derived, not chosen: $2 \leq \ell \leq 20$ at 5%, generated by `scripts/derive_ell_range.py`.³

6 Evidence

Each figure is generated by the script named in its caption and is accompanied by a standalone caption file in `captions/`. This section is itself generated by `scripts/make_figure_pages.py`, so a figure appears here only once it exists in the tree, and what is outstanding is stated rather than omitted.

Outstanding at this release candidate: **F2** truncation convergence, both frames; **F3** angular autocorrelation; **F4** real-space angular correlation; **F5** source decomposition, two frames; **F6** impact of the approximations; **F7** frame specialisation; **F8** coupling schematic. The agreed specification for the full set is `captions/FIGURE-PLAN-v1.0.0.md`.

7 Parameter and numerical evidence

Table 3 lists every number the numerics use. Its *provenance* column distinguishes values taken from the source paper, from standard values that are not, from values derived here, and from values stipulated for this release. The distinction matters: $\Omega_\gamma h^2$ is a standard value and not one the antecedent supplies, and it enters only through $R(\eta_*)$ and hence the sound speed.

Table 3: Every number the numerics use. *Provenance* distinguishes values taken from the source paper, standard values that are not, values derived here, and values stipulated for this release.

Symbol	Code	Value	Provenance	Set in
k_{com}	K_COM	1.0	stipulated	<code>scripts/figure_f1_appendix_f.py</code>
ℓ_{max}	ELL_MAX	80	stipulated	<code>scripts/figure_f1_appendix_f.py</code>
ℓ_{compared}	ELL_COMPARED	25	stipulated	<code>scripts/figure_f1_appendix_f.py</code>
rtol, atol	<code>rtol, atol</code>	$10^{-10}, 10^{-12}$	stipulated	<code>functions/harmonics/free_streaming.py</code>

³That derivation uses the Bessel identity of §8.3, which requires $[(m/2)!]^2$ in its denominator. As printed it is exact where $\Gamma(m/2 + 1) = 1$, so the large-scale result (201) is unaffected; the horizon-scale normalisation (204) and the D_ℓ of (205) are. A misprint, corrected silently and recorded.

Symbol	Code	Value	Provenance	Set in
z_*	Z_STAR	1100	standard	scripts/derive_ell_range.py
h	H	0.5	AnnalsII §8.3.2	scripts/derive_ell_range.py
Ω_B	OMEGA_B	0.05	AnnalsII §8.3.2	scripts/derive_ell_range.py
$\Omega_\gamma h^2$	OMEGA_GAMMA_H2	2.47×10^{-5}	standard	scripts/derive_ell_range.py
tolerance	ADOPTED_TOLERANCE	5%	derived	scripts/derive_ell_range.py
acceptance threshold	--	10^{-11}	stipulated	scripts/figure_f1_appendix_f.py

8 Numerical resolution and reproducibility

The hierarchy is integrated with an explicit Runge–Kutta method of order 8 (DOP853) at relative tolerance 10^{-10} and absolute tolerance 10^{-12} , truncated at $\ell_{\max} = 80$ with comparisons restricted to $\ell \leq 25$ so that reflection from the truncation does not contaminate them. The acceptance threshold used in the figures, 10^{-11} , sits an order above the integrator floor.

The release route is

```
python scripts/run_all.py --strict
```

which regenerates every output, figure and table, recomputes the SHA-256 manifests, and fails if any artefact differs from the recorded hash. `FILE-MANIFEST-SHA256.txt` is the release fingerprint; `IMMUTABLE-MANIFEST-SHA256.txt` covers frozen reference material, where any change is release-blocking. The corrections record is append-guarded rather than frozen, because it must take new findings: appending is permitted and rewriting recorded bytes is not.

Weights are carried as exact rationals rather than floating-point numbers wherever a sign or an ℓ -weight is asserted.

9 Interpretive limits

This is an analytical reproducibility bundle for the associated paper and its supplement. It is not an empirical calibration, not a parameter-estimation pipeline, and not a Boltzmann code for general use. No new effect is claimed and none is reported.

The release does **not** produce a Λ CDM angular power spectrum, and the distinction is not a matter of effort. *Annals II*’s own §8.3.2 model is standard CDM — $h = 0.5$, $\Omega_B = 0.05$, flat and matter dominated — and its C_ℓ is a closed-form Sachs–Wolfe result on large scales with the transfer function set to unity. A Λ CDM spectrum requires a recombination history, tight coupling, diffusion damping and an integrated Sachs–Wolfe contribution from Λ : a full Boltzmann solve, not an extension of the analytic reconstruction. It is the **v2.0.0** deliverable, where *calibration* means calibrating the solver against standard results.

Two statements must travel together, and neither stands alone: the nonlinear contribution $\delta\tau_{NL}$ **matters** — it is not negligible and not an artefact — and it **adds nothing**, redistributing power rather than creating it, because $\delta\tau_{NL} \propto \Phi\Phi$ makes its contribution to C_ℓ quartic in Φ while the linear C_ℓ is quadratic.

F1 The covariant free-streaming recursion against three external hierarchies

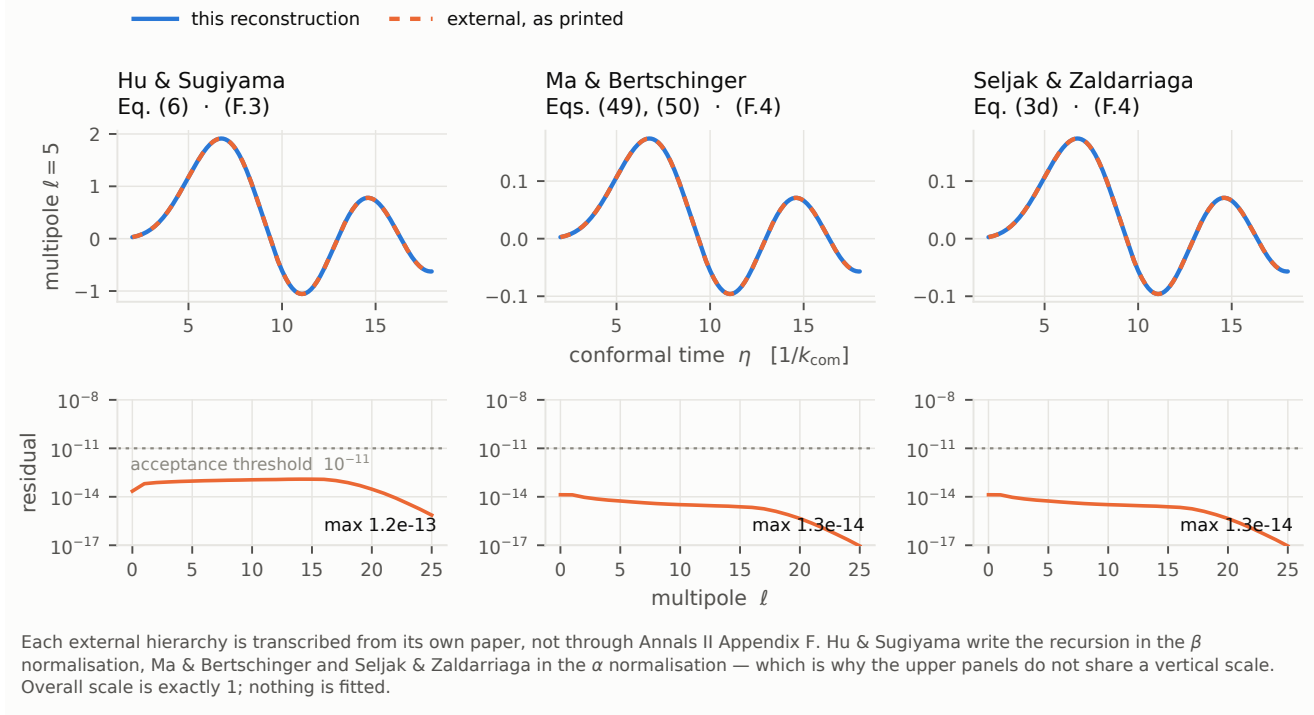


Figure 1: **F1 — The covariant free-streaming recursion against three external hierarchies.** The covariant free-streaming recursion of *Annals II* Appendix F, integrated from a unit monopole, against three external treatments of the same hierarchy. Each is transcribed from its own paper, never through Appendix F. **Upper panels, context:** the reconstruction and the external form at $\ell = 5$; they coincide. The panels do *not* share a vertical scale, because two normalisations are on display. **Lower panels, the claim:** $\max_{\eta} |\text{external} - \text{reconstruction}|$ against ℓ , with the acceptance threshold marked. Hu & Sugiyama write the recursion in the β normalisation and Ma & Bertschinger and Seljak & Zaldarriaga in the α normalisation, which is why Appendix F prints both (F.3) and (F.4); a basis phase leaking into the couplings would break at least one match. Overall scale is exactly 1.0000000000 — nothing is fitted. Three panels, not four: Wilson is pre-arXiv and the citation is ambiguous.

Evidence category: cross-check. Generated by scripts/figure_f1_appendix_f.py. Standalone caption: captions/F1-appendix-f.md.